

Week 2 Lecture Notes: Superposition and Quantum Gates

Cornell Quantum Computing Association - Fundamentals

1 Review and Building Forward (5 minutes)

Welcome back! Last week we took our first steps into the quantum world by understanding how quantum coins differ from classical coins. We discovered that while classical coins must be definitively heads or tails, quantum coins can exist in superposition - simultaneously both heads and tails until measured. We also saw how this leads to exponential advantages: n quantum coins can represent all 2^n classical possibilities at once, compared to classical systems that need 2^n separate computers.

Today we're going to dive much deeper into the mathematical machinery that makes quantum computing work. We'll explore superposition with mathematical precision, learn about the full toolkit of quantum gates that let us manipulate quantum information, and start building the foundation for quantum algorithms. By the end of today, you'll understand how to design quantum circuits and predict their behavior with mathematical accuracy.

The journey today will take us through three main areas: first, we'll develop a complete mathematical understanding of superposition using probability amplitudes and complex numbers. Then we'll explore the Bloch sphere - a beautiful geometric way to visualize quantum states. Finally, we'll learn about quantum gates systematically, seeing how they work both mathematically and geometrically.

2 Deep Dive into Superposition (15 minutes)

2.1 Mathematical Representation of Quantum States

Last week we introduced the idea that a qubit can be in superposition, written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Now let's understand this with full mathematical rigor. The coefficients α and β are called probability amplitudes, and they can be complex numbers. For our purposes, we can write them in the form: $\alpha = a + bi$, $\beta = c + di$

where a , b , c , and d are real numbers, and i is the imaginary unit with $i^2 = -1$.

When we measure a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we get:

- Outcome $|0\rangle$ with probability $|\alpha|^2 = a^2 + b^2$
- Outcome $|1\rangle$ with probability $|\beta|^2 = c^2 + d^2$

This is the **Born rule** - one of the fundamental postulates of quantum mechanics. The key insight is that probabilities come from the squared magnitudes of amplitudes, not the amplitudes themselves.

2.2 The Normalization Condition

Since we must get some outcome when we measure, the probabilities must sum to 1: $|\alpha|^2 + |\beta|^2 = 1$

This is called the **normalization condition**. It constrains which quantum states are physically possible. For example:

- $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ is valid since $|\frac{1}{\sqrt{2}}|^2 + |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} + \frac{1}{2} = 1$
- $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$ is not valid since $|\frac{1}{2}|^2 + |\frac{1}{2}|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \neq 1$

2.3 Complex Amplitudes and Quantum Interference

Why do we need complex numbers? The complex nature of quantum amplitudes is crucial for interference effects. Consider these two states: $|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ $|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{-1}{\sqrt{2}}|1\rangle$

Both states give the same measurement probabilities: 50% chance of $|0\rangle$ and 50% chance of $|1\rangle$. But the minus sign in $|\psi_2\rangle$ represents a crucial difference that becomes important when we apply quantum gates.

We can be even more explicit about superposition. A general qubit state is: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = (a + bi)|0\rangle + (c + di)|1\rangle$

This means our qubit simultaneously exists in both the $|0\rangle$ and $|1\rangle$ states, with the complex amplitudes determining both the probability of measuring each outcome and how the states interfere with each other during quantum operations.

2.4 Multiple Qubit Systems

When we have multiple qubits, the state space grows exponentially. For two qubits, the general state is:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

with the normalization condition:

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

For n qubits, we need 2^n complex amplitudes to specify the state completely. This exponential scaling is both the source of quantum computing's power and the reason quantum systems are so hard to simulate classically.

3 The Bloch Sphere: Visualizing Quantum States (10 minutes)

3.1 Geometric Representation

The Bloch sphere provides a beautiful geometric way to visualize single qubit states. It's a unit sphere where every point on the surface represents a possible qubit state.

The mapping works as follows:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

corresponds to the point on the Bloch sphere with spherical coordinates (θ, ϕ) . Here:

- θ is the polar angle (0 to π)

- ϕ is the azimuthal angle (0 to 2π)

Key positions on the Bloch sphere:

- North pole $(0, 0)$: $|0\rangle$
- South pole $(\pi, 0)$: $|1\rangle$
- Point $(\pi/2, 0)$: $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ (the $|+\rangle$ state)
- Point $(\pi/2, \pi)$: $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ (the $|-\rangle$ state)

3.2 Why the Bloch Sphere is Useful

The Bloch sphere helps us understand several important concepts:

1. **Measurement probabilities:** The probability of measuring $|0\rangle$ is $\cos^2(\theta/2)$, which depends only on how "high" the point is on the sphere.
2. **Quantum gates as rotations:** Every single-qubit quantum gate corresponds to a rotation of the Bloch sphere.
3. **Phase relationships:** Points on the same latitude have the same measurement probabilities but different phases.
4. **Antipodal points:** Opposite points on the sphere represent orthogonal states that give opposite measurement outcomes with certainty.

3.3 Limitations of the Bloch Sphere

The Bloch sphere only works for single qubits. For multiple qubits, we need higher-dimensional spaces that can't be easily visualized. This is one reason why quantum mechanics becomes harder to understand as we scale up - our geometric intuition breaks down.

However, the Bloch sphere remains incredibly valuable for understanding single-qubit operations and building intuition about quantum states and transformations.

4 Essential Quantum Gates (20 minutes)

Now we'll systematically learn the fundamental quantum gates that form the building blocks of quantum algorithms. Each gate can be understood as a matrix operation and as a rotation on the Bloch sphere.

4.1 The Pauli Gates

The three Pauli gates are fundamental single-qubit operations:

X Gate (Quantum NOT): $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Action: $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$

On the Bloch sphere: 180 deg rotation around the X-axis

Y Gate: $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

Action: $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$

On the Bloch sphere: 180 deg rotation around the Y-axis

Z Gate: $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Action: $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$

On the Bloch sphere: 180 deg rotation around the Z-axis

The Z gate is particularly important because it adds a phase to the $|1\rangle$ component without changing measurement probabilities.

4.2 The Hadamard Gate

We introduced the Hadamard gate last week, but let's understand it more deeply:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Actions:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \quad (1)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle \quad (2)$$

The Hadamard gate creates superposition from classical states and can also "unmix" certain superposition states back to classical states. On the Bloch sphere, it's a 180 deg rotation around an axis halfway between X and Z.

4.3 Phase Gates

Phase gates modify the phase of quantum states without changing measurement probabilities:

S Gate (Phase gate): $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

Action: $S|0\rangle = |0\rangle$ and $S|1\rangle = i|1\rangle$

T Gate ($\pi/8$ gate): $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

Action: $T|0\rangle = |0\rangle$ and $T|1\rangle = e^{i\pi/4}|1\rangle$

These gates are crucial for creating the interference effects needed in quantum algorithms.

4.4 Gate Composition and Circuits

Quantum gates can be combined in sequence to create more complex operations. When gates are applied in sequence, we multiply their matrices:

If we apply gate A followed by gate B , the combined effect is:

$$BA$$

Note the reverse order - this is because we read quantum circuits from left to right, but matrix multiplication is right to left.

Example: What happens when we apply H, then Z, then H again?

$$H \cdot Z \cdot H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$$

So $HZH = X$! This is an example of how gate sequences can implement other gates.

5 Quantum Circuit Notation (8 minutes)

5.1 Reading Quantum Circuits

Quantum circuits are read from left to right, with time flowing left to right. Each horizontal line represents a qubit, and gates are represented as boxes or symbols on these lines.

Basic symbols:

- X gate: $\oplus$ (circle with plus)
- H gate: [H] (box with H)
- Z gate: [Z] (box with Z)
- Measurement: meter symbol

5.2 Circuit Construction Principles

When building quantum circuits, remember:

- **Initialization:** Qubits start in $|0\rangle$ unless specified otherwise.
- **Gate application:** Gates modify qubit states according to their matrix representations.
- **Measurement:** Converts quantum states to classical outcomes.
- **No cloning:** You cannot copy arbitrary quantum states.
- **Reversibility:** All quantum gates (except measurement) are reversible.

5.3 Analyzing Circuit Behavior

To analyze a quantum circuit:

- Start with the initial state (usually $|00\dots 0\rangle$)
- Apply each gate from left to right
- Track the quantum state after each gate
- Calculate final measurement probabilities
- Initial: $|0\rangle$
- Apply H: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- Apply Z: $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- Apply H: $|1\rangle$

So the circuit HZH converts $|0\rangle$ to $|1\rangle$ - it implements the X gate!

6 Multiple Qubits and Gate Applications (10 minutes)

6.1 Product States vs Entangled States

For multiple qubits, we can have different types of states:

Product states (separable): Can be written as a product of single-qubit states

$$|\psi\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Entangled states: Cannot be written as products of single-qubit states

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This distinction will become crucial next week when we explore entanglement in detail.

6.2 Single-Qubit Gates on Multiple-Qubit Systems

When we apply a single-qubit gate to one qubit in a multi-qubit system, it acts on that qubit while leaving others unchanged. Mathematically, this is described using the tensor product:

To apply X gate to the first qubit of a two-qubit system:

$$X \otimes I = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

6.3 Interference in Multiple-Qubit Systems

With multiple qubits in superposition, we can create interference effects between different computational paths. This is the foundation of quantum algorithms.

Consider two qubits, both in superposition after applying H gates:

$$|++\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Different quantum operations can cause these amplitudes to interfere constructively or destructively, allowing us to amplify desired outcomes and suppress undesired ones.

7 Building Toward Quantum Algorithms (7 minutes)

7.1 The Recipe for Quantum Advantage

Now we have the essential ingredients for quantum algorithms:

- **Superposition creation:** Use H gates to put qubits into superposition
- **Amplitude manipulation:** Use various gates to modify probability amplitudes
- **Interference engineering:** Design gate sequences so wrong answers cancel and right answers amplify
- **Measurement:** Extract classical information from quantum states

Next week, we'll see how entanglement adds another powerful tool to this toolkit.

7.2 Preview: Deutsch's Algorithm

In two weeks, we'll implement Deutsch's algorithm, which uses these principles:

- Start with superposition (H gates)
- Apply an oracle that encodes the problem
- Use interference to distinguish function types
- Measure to get the answer

The algorithm solves a problem with one quantum query that requires two classical queries - our first example of quantum advantage.

7.3 What We've Accomplished

Today we've built a complete mathematical foundation for single-qubit quantum computing:

- Precise understanding of superposition and probability amplitudes
- Geometric visualization through the Bloch sphere
- Systematic knowledge of fundamental quantum gates
- Ability to analyze and design quantum circuits

These tools will serve us throughout the rest of the course as we explore entanglement, implement quantum algorithms, and eventually program real quantum computers.